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Artificial intelligence application and high-performance work systems in the manufacturing sector: a moderated-mediating model

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Abstract
This empirical investigation examines the complex dynamics between Artificial Intelligence (AI), Potential Development (PD), Training Initiatives (TI), and High-Performance Work Systems (HPWS) within manufacturing firms to gain valuable insights into how AI technologies influence high-performance work systems through employee development and training. Using a purposive sampling technique, around two hundred employees from twenty-four manufacturing firms in the textile, automotive, steel, and pharmaceutical sectors participated in the self-administered survey. The empirical analysis of the data sets was conducted using the PLS-SEM approach. This result demonstrated positive associations between AI, PD, and HPWS, emphasizing the key role of AI in supporting employee development and improving high-performance work systems. Furthermore, training's amplification effect on the relation between artificial intelligence and professional development highlighted the significance of employees' upskilling for AI integration. Conversely, the mediating role of PD between AI adoption and HPWS effectiveness highlighted the significant role of employee professional development in achieving HPWS through AI integration within the systems. The study offered insight into the mediation of PD between AI and HPWS effectiveness, emphasizing its centrality in translating AI-driven advances into tangible organizational outcomes. The study findings have significant ramifications for both theory and practice. Theoretically, this research adds to an evolving dialogue surrounding AI's effects on HR practices and organizational outcomes; practically speaking, organizations can utilize this research's insights in strategically integrating AI technologies, designing tailored training programs for their employees, and creating an environment conducive to ongoing employee development.

Keywords Applied artificial intelligence · Potential development · Training initiatives · High-performance work systems · Manufacturing sector

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1 Introduction

Recent technological advancements have resulted in the rapid implementation of Artificial Intelligence (AI), revolutionizing how businesses operate and make decisions (Alenzi 2023). Artificial Intelligence (AAI) has unlocked new possibilities and challenges across various sectors, shifting industries' landscapes and altering traditional practices (Ari et al. 2020). Manufacturing sectors – particularly manufacturing operations that play a vital role in economies – have seen significant adoption of AAI for operational efficiency, productivity, and competitiveness enhancement (Ari et al. 2020).

AAI holds great promise in manufacturing for optimizing processes, increasing efficiency, and elevating overall competitiveness - yet its transformative power extends far beyond operations enhancements to influence human resources management practices and organizational dynamics (Binsaeed et al. 2023). AAI plays an integral part in HPWS design/implementation, specifically, the impact AAI has had on improving employee skills, motivation levels, and organizational performance (Adil and Ab Hamid 2020). AAI introduces advanced technologies that optimize processes, enhance decision-making processes, and drive efficiency across various domains, yet how AAI influences HPWS practices remains unclear (Behravesh et al. 2020).

As technology innovation rapidly evolves, human expertise and artificial intelligence (AI) have become the subject of significant research interest from scholars, practitioners, and policymakers alike (Das 2022). To date, AAI and HPWS interactions remain unclear due to their complex interaction, though reverberated through potential development and training (Alenzi 2023). Potential development becomes central when AAI augments workforce capabilities through aspirations-inducing technology (Zhou and Zou 2023). Additionally, training plays a vital role in the dynamics between artificial intelligence integration and potential development by amplifying any positive effects or mitigating the risks (Kaushik and Mukherjee 2022).

This research seeks a holistic view of how applied artificial intelligence, employee potential development, and training influence high-performance work systems (Zhou and Zou 2023). Furthermore, by investigating the mediating role of employee professional development, this study offers an insightful look into how AAI technology ties in with human capital development goals, ultimately creating High-Performance Work Systems (Zhou and Zou 2023). These insights can assist organizations in harnessing AAI to optimize operational processes while building high-performing workforces (Ari et al. 2020).

The current study aims to fill a knowledge void between applied artificial intelligence, employee potential development, and high-performance work systems for effective human resource management (Singh et al. 2020; Li et al. 2022). By unraveling intricate threads linking these components, this research contributes to broader discussions around organizational performance while opening doors for informed decision-making during an age of technological innovation (Narbariya et al. 2022). It will explain how AAI marries High-Performance Work Systems (HPWS) (Das 2022).

Pakistan's manufacturing sector, an essential contributor to the national economy, stands to take advantage of Artificial Intelligence to remain globally competitive (Ari et al. 2020). With machine learning, robotics, and data analytics becoming more pervasive across industries and organizations (Gittell and Ali 2021), understanding their effects on employee engagement, skills development, and overall performance, as well as mediating and mod-

erating factors remains an under-explored area, particularly within Pakistani manufacturing (Pap et al. 2022).

The motivation to interrogate AAI/HPWS dynamics in Pakistan's manufacturing sector transcends the traditional boundaries (Behravesh et al. 2020). Much research has been conducted on AAI and HPWS individually, but their joint impact on Pakistan's manufacturing sector is yet to be explored (Pap et al. 2022). Organizations cannot fully optimize their artificial intelligence strategies for high-performance work systems unless not explored from training and development perspectives (Mohsin 2023). This comprehensive investigation enriches academic discourse and offers actionable insights for manufacturing practitioners navigating technology, human capital development, and performance enhancement (Arshad et al. 2022).

This study examines the interplay among applied AI, employee potential development, training initiatives, and high-performance work systems. It seeks to determine the impact of applied artificial intelligence (AAI) on high-performance work systems within Pakistan's manufacturing sector, with potential development as an intermediary and training as a moderator (Gittell and Ali 2021). The study will answer our research questions: Does applied AI positively impact employee potential development and high-performance work systems? Does employee potential development mediate the impact of applied AI on high-performance work systems? Do training initiatives moderate the relationship between applied AI and employee potential development? The findings will contribute to the knowledge domain by uncovering hidden dynamics among AAI, employee potential development, training programs, and implementation of high-performance work systems (Alam and Lin 2022).

In the next section, the paper outlines the study's theoretical background that led to developing the research framework and hypotheses. After this, research methods and data collection approaches are highlighted with a detailed account of data testing and results. The findings are discussed, and the study concludes with implications, limitations, and future directions.

2 Theoretical background

2.1 Applied Artificial Intelligence (AAI)

Applied Artificial Intelligence (AAI) represents a revolutionary shift in how organizations operate and innovate across various sectors (Alenzi 2023). AAI refers to the strategic integration of advanced Artificial Intelligence technologies into different aspects of an organization, such as business processes, decision-making frameworks, and complex problem-solving scenarios - often by using cutting-edge algorithms, machine learning models, and data-driven insights (Ihsan et al. 2020). Organizations use AAI for automation, data analytics, and predictive modeling purposes. Scholars and practitioners have highlighted its revolutionary potential as it optimizes operations while driving innovation within industries; however, limited research has been undertaken on how AAI influences organizational processes and human resource practices (Arshad et al. 2022).

Studies on artificial intelligence implementation have focused primarily on its technical dimensions with a marginal focus on human resources practices and management. AI can profoundly impact HR practices beyond operational enhancement (Ihsan et al. 2020).

Integrating AAI into organizational frameworks requires an intricate knowledge of how it affects employee roles, skill requirements, and career trajectories. Once organizations embrace AAI, it becomes crucial to examine its implications for workforce development and enhancement and for designing High-Performance Work Systems (HPWS) (Das 2022). As organizations adapt to an ever-evolving work landscape, they must emphasize potential development equally as they prepare a workforce ready for success in an age of artificial intelligence (Jani & Moduli, 2021).

2.2 High-Performance Work Systems (HPWS)

HPWS is an umbrella encapsulating a set of human resource practices designed to maximize employee capabilities, align individual and organizational goals and enhance overall performance (Pap et al. 2022). Such practices include selective hiring, comprehensive training programs, performance-based compensation arrangements, and opportunities for skill utilization. HPWS has been linked with improved employee satisfaction, productivity, and organizational outcomes, yet the influence of AAI on its effectiveness remains relatively unexplored (Sassanelli and Terzi 2023).

High-Performance Work Systems (HPWS) represent an integrated and holistic approach to human resource management that seeks to create an ideal working environment where employees can maximize their skills and the organization's performance (Ari et al. 2020). This multifaceted framework comprises practices, policies, and strategies that connect individual motivation with organizational objectives (Das 2022). Empirical research underscores the positive effect of HPWS practices on various dimensions of organizational performance. According to research by Pap et al. (2022), organizations adopting HPWS practices experience higher employee satisfaction rates, lower turnover rates, and enhanced financial results (Ari et al. 2020).

Furthermore, HPWS has been linked with increased employee motivation, commitment, and job performance. While these benefits of HPWS are well-known, the role of Artificial Intelligence and its integration in HPWS remains intriguing yet underexplored. Integrating Artificial Intelligence into organizational processes creates new dynamics that intersect with HPWS practices (Ari et al. 2020). For instance, AAI technologies might streamline recruitment by automating initial screening or enhancing training programs through personalized learning algorithms (Alenzi 2023).

However, to create an atmosphere conducive to learning and potential development, AAI and HPWS could provide potential synergies. Few studies have systematically explored how AI interacts with HPWS components (Ari et al. 2020). Scholarly investigations have explored its influence on job design, candidates' recruitment and selection, data analytics' role in performance evaluation, and the effects of AAI-powered decision-making on employee motivation (Ihsan et al. 2020). In the future, as organizations increase the use of AAI to boost their productivity, it is vital to understand its effects on HPWS - such as potential enhancements, challenges, and adaptations - before moving forward with its blanket adoption in all organizational processes. Exploring this complex relationship will offer detailed insight into how AAI ties in with and can potentially transform HPWS positively, shaping work in today's rapidly developing technological milieu (Ayestaran Echeverria et al., 2022).

2.3 Employee Training

Training is a cornerstone of skill development and knowledge enhancement. Effective training programs create capable and adaptable employees who can navigate today's fast-changing business landscape (Zhou and Zou 2023). Literature highlights training's role in increasing employees' performance while creating an environment of continuous learning (Desta et al. 2022).

Training is integral to organizational development, providing employees with the necessary skills, knowledge, and competencies to succeed. Training programs have become indispensable in today's complex and ever-evolving business environment. They bolster workforce capabilities and foster adaptability and innovation (Ihsan et al. 2020). Well-structured training initiatives allow employees to perform their current tasks proficiently and equip them to face any upcoming challenges or seize new opportunities (Zhou and Zou 2023). Training increases organizational agility by equipping employees with shifting industry landscapes, technological innovations, and evolving customer expectations (Faisal, 2022).

The literature emphasizes the significance of training programs in improving employee potential and performance. Comprehensive training initiatives enhance job mastery, reduce errors, and increase efficiency and consistency in output (Ari et al. 2020). Furthermore, training initiatives are crucial in creating an environment of continuous learning. When employees have opportunities to advance their knowledge and abilities, they are more likely to participate in self-directed learning and help create an atmosphere conducive to innovation and adaptability (Ali 2021).

Research by Pap et al. (2022) highlights the vital link between training initiatives, employee performance, and job satisfaction. Their study shows how well-designed training interventions improve job performance and satisfaction levels. While Applied Artificial Intelligence (AAI) and Potential Development have transformative power, the effect of training on their relationship remains an interesting avenue to explore (Alam and Lin 2022).

2.4 Applied Artificial Intelligence, Employee Potential Development, & High-Performance Work System

Artificial Intelligence (AAI) has quickly emerged as a transformative force within the manufacturing sector, as its rapid evolution warrants (Das 2024). AAI encompasses various AI technologies–ranging from machine learning algorithms and natural language processing–which are harnessed to streamline operations, enhance customer experiences, and drive data-informed decision-making in manufacturing firms (Al-Taweel 2021). While AAI's ability to reshape manufacturing operations is widely acknowledged, its specific impact on employee's potential development within this sector remains somewhat understudied (Al-Taweel 2021).

Data analysis provided by AAI offers actionable insights for risk evaluation and fraud detection. These innovations go beyond operational efficiency and enhance employee skill enhancement and career advancement (Zhou and Zou 2023). Routine tasks that were once manual are now automated so employees can dedicate more of their efforts toward higher-value activities (Ihsan et al. 2020). Employees interacting with AI systems must adapt their skill sets to function optimally, creating an atmosphere conducive to learning and growth.

A study by Alam and Lin (2022) demonstrated how AAI implementation could encourage skill diversification and continuous learning and contribute to employees' potential development (Gittell and Ali 2021).

AI-powered tools offer employees dynamic platforms to hone their competencies (Ari et al. 2020). For example, employees interacting with customer service tools driven by AI can hone interpersonal skills while engaging in interactions that require human empathy and understanding; similarly, AI data analysis tools give employees access to interpret and leverage insights from vast datasets while honing analytical abilities and decision-making acumen (Ari et al. 2020). Alongside, AAI brings transformative capabilities that add new dimensions to organizational processes and human resource practices (Binsaeed et al. 2023).

The prior literature emphasizes a correlation between employee potential development and HPWS effectiveness (Singh et al. 2020). Organizations that invest in employees' potential often see increased engagement and commitment among staff members, creating an ideal setting for HPWS implementation (Line, 2022). However, more research needs to be conducted to understand its mediating role between AAI and HPWS practices (Line, 2022).

Organizations that invest in nurturing employees' potential see numerous positive effects (Ari et al. 2020). Research highlights a positive correlation between potential development and HPWS practices, as organizations that prioritize potential development experience greater employee engagement, commitment, motivation, and engagement with the workplace culture (Chekalina 2024). Employees who feel that their development is well taken care of by their organization are more likely to commit their energies to HPWS practices (Adil and Ab Hamid 2020).

Potential development provides employees with the tools and confidence needed to excel in their roles, seamlessly fitting with the principles of HPWS (Zhou and Zou 2023). When employees can effectively leverage their enhanced skill sets in collaborative efforts and take ownership over assigned tasks and responsibilities, their involvement becomes critical in collaborative endeavors as they assume ownership of tasks and responsibilities (Pap et al. 2022). Studies by Khan and Liu (2023) illustrate that employee potential development directly contributes to HPWS effectiveness by increasing employee performance and creating an atmosphere of shared purpose among team members (Khan and Liu 2023).

The integration of artificial intelligence (AI) technologies within organizational frameworks has become increasingly prevalent in recent years. However, while the adoption of AI holds significant promise for enhancing productivity and efficiency within organizations, it also brings forth a myriad of implications that must be carefully examined. In this paper, we aim to provide a thorough analysis of the usage of AI in organizations, with a particular focus on its implications for employee productivity (Leng et al. 2024).

To establish a strong theoretical foundation for our analysis, we draw upon the works of prominent researchers such as Davenport (2007) and recent studies by Raish and Krakowski (2021). These scholars have extensively explored the usage of AI in organizations and have shed light on its multifaceted impacts on various organizational processes. However, it is imperative to note that the landscape of AI in organizations is continuously evolving, and recent empirical evidence must be integrated into our analysis to provide a comprehensive understanding of the subject matter (Kazemi and Milani 2024).

2.5 Mediating role of Employee Potential Development

An examination of potential development as an intermediary between Artificial Intelligence (AAI) and High-Performance Work Systems (HPWS) represents an essential pursuit in understanding how AAI transforms organizational performance (Singh et al. 2020). On one side, artificial intelligence can dramatically change how organizations operate, innovate, and engage with their workforce (Singh et al. 2020), and on the other side, potential development is an integral element in achieving organizational excellence (Singh et al. 2020), whereby nurturing employees' skills, inculcating growth mindsets, and providing career trajectories, potential development fosters improved employee engagement, commitment, and motivation levels.

2.6 Moderating role of Training Initiatives

With manufacturing industries at the surge of AI technology adoption to optimize operations and enhance customer experiences, understanding how training can modulate the effect of AI on employee potential development is vital (Behl et al. 2021). An effective training program equips employees with the necessary AI knowledge, instilling confidence in its use and helping them to leverage its transformative potential (Ari et al. 2020). Training links AI's technological integration and employees' potential development (Al-Taweel 2021) and impacts employees' potential development, which indicates the synergistic alignment of training with AI and employees' professional development. An empirical investigation may offer insight into this complex relationship (Zhou and Zou 2023) and help to determine if training can moderate the relationship between artificial intelligence and employee potential development, especially within manufacturing sectors, which remain primarily underexplored (Alenzi 2023).

3 Research Framework & Hypotheses

This research draws upon two established theories of social cognitive and resource-based to understand the complex interactions among Artificial Intelligence (AAI), Potential Development (PD), High-Performance Work Systems (HPWS), and training in the Pakistani manufacturing sector. These theories offer a way of seeing through these relationships and understanding the underlying mechanisms and dynamics governing them.

3.1 Social Cognitive Theory (SCT)

Social Cognitive Theory proposed by Bandura (1986) is based on agentic perspectives that 'people are self-organizing, proactive, self-reflecting, and self-regulating and not just reactive shaped by environmental events or inner forces' (Bandura 2001, p.266). Following the paradigm, Li and colleagues (2022) hold that individuals learn by observing, mimicking, and modeling behaviors demonstrated by others. AAI integration within manufacturing introduces new technological behaviors and practices. When exposed to AAI technologies - like customer service AI or data analytics systems - employees observe them, interact with them, and eventually gain new skills or competencies through observational learning

(Narbariya et al. 2022). This way, they engage in self-regulation and self-efficacy beliefs, which affect their willingness to engage in certain behaviors (Al-Taweel 2021). Employees see the impact of AAI technologies on organizational processes and develop an enhanced sense of self-efficacy when using them - leading them to take proactive steps towards further developing themselves by finding opportunities to acquire skills or explore potential (Ihsan et al. 2020).

3.2 Resource-Based View (RBV)

Barney's (1991) Resource-Based View Theory holds that firms gain a competitive advantage by strategically exploiting unique and valuable resources. It states that firms develop competitive advantage by acquiring, developing, combining, and deploying their physical, human, and organizational resources in a way that adds unique value and is difficult to imitate by competitors (Colbert 2004, p. 343). However, resource-based arguments are primarily rooted in human resources knowledge, skills, behaviors, and competencies built over time and inimitable (Barney 1992). Hence, RBV is strongly connected to HPWS practices and leverages employee skills, competencies, and potential development as resources contributing to organizational performance. AAI enhances these intangible assets by influencing potential development, ultimately improving their organizational arsenal of human capital resources and facilitating HPWS practices (Mohsin 2023).

RBV suggests that firms can enhance the value of their resources through strategic investments (Ayestaran et al., 2022). Training is integral in moderating the relationship between applied AI and potential development by increasing employees' potential development value. Effective training programs enable employees to use AAI technologies more effectively, thus increasing the resource value of potential development, further intensifying its effects, and leading to continuous enhancement (Jani and Muduli 2021).

Based on the premise of the SCT and RBV, the study (see research framework; Fig. 1) examines if applied artificial intelligence (AAI) affects high-performance work systems (HPWS) practices directly or through employee potential development (PD) (Mohsin 2023).

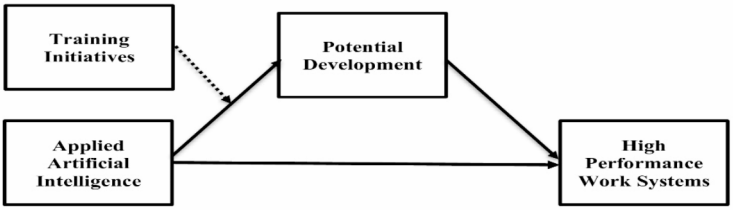

Fig. 1 Research Model

Additionally, as previous studies suggested to explore the interrelationships among applied artificial intelligence, training, and high-performance work systems in the manufacturing industry (Zhou and Zou 2023), the current study also assesses if the training initiatives moderate the impact of AI application on employees' professional development, subsequently effecting the high-performance work systems. The hypothesized relationship between the study constructs will help us unravel complex mechanisms underlying technology adoption, employee training, their potential development, and overall performance enhancement (Singh et al. 2020). The study hypotheses that:

H1. There is a significant positive relationship between applied artificial intelligence and potential development.

H2. There is a significant positive relationship between applied artificial intelligence and High-Performance Work Systems.

H3. There is a significant positive relationship between the potential development and High-Performance Work Systems.

H4. Potential development mediates the relationship between applied artificial intelligence and High-Performance Work Systems.

H5. Training initiatives moderate the relationship between applied artificial intelligence and potential development.

In essence, the research framework determines (a) the significant positive relationships among the study constructs of AAI, PD, and HPWS; (b) the mediating role of PD between AAI and HPWS; and (c) the moderating role of training between AAI and PD relationship. The study constructs have been operationalized in the research as follows:

Applied Artificial Intelligence (AI) In this study, the 'applied artificial intelligence' construct is operationalized as the utilization of AI technologies and tools to address challenges and solve issues in the manufacturing sector (Ayestarán et al. 2022).

Professional Development In this study, the 'professional development' variable is operationalized as structured learning and skill enhancement programs or informal practices aimed at employees' development beyond that derived from their initial training (Collin et al. 2012).

Training Initiatives The study operationalizes 'Training initiatives' as organized programs designed to provide employees with the required information and new skills to achieve organizational goals (Truitt 2011).

High-Performance Work Systems The study operationalizes the high-performance work systems variable as an integrated approach to HRM that seeks to create an ideal working environment where employees can maximize their skillset and the organization's performance (Ari et al. 2020).

4 Research Methodology

The current study employed a quantitative empirical approach to explore the hypothesized relationships among study endogenous and exogenous constructs of Applied Artificial Intelligence (AAI), Potential Development (PD), High-Performance Work Systems (HPWS), and training (Tr) within Pakistan's manufacturing sector. The structured data collection was conducted, followed by analysis, interpretation, and presentation to meet research objectives and hypotheses. The cross-sectional study took a snapshot of relationships among

variables at one specific moment, which allowed the examination of proposed relationships and the assessment of mediation and moderation effects.

4.1 Data Collection

The exploratory study delved into the impacts of AI adoption on employee productivity within organizational settings. We rigorously examined the complex relationships at play by employing Partial Least Squares Structural Equation Modeling (PLS-SEM) as our primary analytical tool. Rigorous sampling techniques and diverse data collection methods, including surveys and interviews, ensured the reliability of our findings. Mitigating common method bias and assessing model fit were integral steps in our analysis, allowing us to draw robust conclusions. Explicit consideration of moderating effects shed light on the nuanced nature of AI's influence in different organizational contexts (Plathottam et al. 2023). Our research contributes valuable insights to the ongoing discourse on AI adoption and its implications for organizational management.

The self-administered survey was gathered through an online survey using the structured questionnaire to collect the data from the employees and managers working in the manufacturing sector in Pakistan. The manufacturing sector is an ideal setting to study the hypothesized relationships due to its rapid advancements in AAI adoption rates and its understanding of the significance of nurturing a skilled and learned workforce within high-performance contexts (Li et al. 2022). The purposive sampling technique was used to collect the data from the employees working in different automotive, pharmaceutical, textile, and steel industries. Industry associations, government records, and publicly available sources were used to identify the targeted firms. An array of manufacturing industries in Pakistan, each one boasting its specialization, were recruited for the study. The firms were chosen from various cities around Pakistan to ensure an even representation. Within the textile industry - famous for cotton yarn production, fabrics, garments, and home textiles production – ten firms participated based in the cities of Karachi, Faisalabad, Lahore, and Sialkot.

Eight firms were recruited from Lahore, Karachi, and Faisalabad in this study of the automotive industry - including the production of cars, motorcycles, commercial vehicles, and auto parts. Next, from the steel industry - known for producing bars, sheets, and, from the steel industry - known for producing bars, sheets, coils, and pipes – six firms from Karachi, Lahore, and Islamabad participated in the study. The diversity of the firms with different products and geographic locations enriched the study findings, which are easily generalizable across the sectors and regions.

Once potential participating firms were identified, their respective HR managers were contacted; the study objectives were shared with them, along with the guidelines for filling out the survey. After the confirmation of participation in the study, a link to the survey was sent out among employees across various hierarchies in the selected firms. The anonymity of responses was confirmed, and a timeline of two weeks was given to fill out the survey forms.

The survey included the items to measure constructs of AAI adoption, Potential Development, HPWS practices, and Training initiatives taken by the firms, using a five-point Likert scale, allowing measurement of workforce perceptions and work experiences. The survey was developed in English, keeping in view a high understanding of the English language among the Pakistani workforces.

The inclusion and exclusion criteria were used to ensure the consistency, reliability, and uniformity of the study population. As discussed earlier, the inclusion criteria focused on selecting large and medium sized firms from the automotive, pharmaceutical, textile, and steel industries within the manufacturing sector due to their significant advancements in AI adoption and their relevance to the research objectives, from various cities across Pakistan, including Karachi, Lahore, Faisalabad, Sialkot, and Islamabad to ensure comprehensive geographic representation. The survey targeted employees across various hierarchies within the selected firms, including managers, engineers, technicians, and other staff involved in or affected by AI applications in their daily operations. The exclusion criteria ensured to focus on the manufacturing sector using AI applications only. Firms outside the manufacturing sector, such as service industry, agriculture, or construction, were excluded. Small enterprises with fewer than 50 employees were also excluded due to potentially limited AI adoption and different operational dynamics compared to larger firms. Additionally, firms outside Pakistan's major industrial cities were excluded to focus on regions with higher concentrations of manufacturing activities and AI adoption rates. To ensure data completeness and reliability, firms or respondents with incomplete surveys or those who did not respond within the given two-week timeline were excluded.

Several procedural and statistical remedies were employed to address common method bias. During the survey design phase, we assured the respondents of the anonymity and confidentiality of their responses to reduce evaluation apprehension. Furthermore, items measuring different constructs were mixed within the questionnaire to reduce the possibility of respondents guessing the relationships among them. Post-data collection, Harman's single-factor test was conducted to assess the presence of common method bias. The results indicated that common method bias was not a significant concern in our study.

Next, an ad hoc power analysis was conducted to estimate the required sample size. Using G*Power software, we determined that a minimum sample size of 200 respondents was needed to achieve a power of 0.80, an effect size of 0.15, and a significance level of 0.05 for our PLS-SEM analysis. This estimation ensured that our study had sufficient statistical power to detect meaningful relationships among the variables.

5 Analyses & Findings

5.1 Measurement Models & Assessments

The scales used to measure the study constructs were adopted from existing studies. HPWS construct was measured using the twenty items developed by the researchers. The respondents indicated their level of agreement on the Likert Scale from 1 (strongly disagree) to 5 (strongly agree).

Likewise, five items were adapted from Gonzalez Canovas (2020) for measuring PD. A five-point Likert scale assessing the frequency of experiences (1 "Never", 2 "Sometimes", 3 "Normally", 4 "Almost Always" " and 5 "Always") was used. Training initiative was measured using the five items adopted from Kambur and Akar (2021) - a five-point Likert scale (ranging from 1-strongly agree to disagree 5-strongly)- to determine the training initiatives taken by the firm to equip their employees. Artificial Intelligence (AI) was measured using the ten items adopted from Kambur and Akar (2021). The respondents marked their agree-

ment or disagreement with statements using a 5-point Likert scale (from 1-strongly agree to disagree 5-strongly) to measure employees' perceptions of AI integration within their respective organizations.

The reliability and validity of the constructs used in this study are critical to the robustness of our findings. As indicated in Table 1, Cronbach's Alpha values for the constructs, including High-Performance Work Systems, Potential Development, Training Initiatives, and Artificial Intelligence, are well above the acceptable threshold of 0.70 (Hair et al. 2011). This demonstrates the internal consistency and reliability of our measurement scales. Additionally, each construct's Average Variance Extracted (AVE) values exceeded the recommended threshold of 0.50 (Fornell and Larcker 1981), confirming convergent validity. These findings (see Table 1) provide confidence in our constructs' measurement accuracy and reliability. To address multicollinearity concerns, we assessed the Variance Inflation Factor (VIF) values for our constructs. VIF values below 5 indicate that multicollinearity is not a significant concern (Hair et al. 2011). Our analysis revealed VIF values (Table 1) well below this threshold, signifying that our constructs are not highly correlated with each other. This supports the independence of our predictor variables. The reliability analysis was conducted to examine the consistency and dependability of the measurement scales employed. Cronbach's alpha coefficients of the study constructs (Table 1) indicated the internal reliability of the variables considered. Notably, the High-Performance Work Systems construct reported Cronbach's alpha score of 0.86, indicating high internal consistency. Similarly, the Potential Development construct demonstrated high reliability with a Cronbach's alpha of 0.81. Also, the training construct had a Cronbach's alpha of 0.82 for acceptable internal consistency. Finally, the Artificial Intelligence construct demonstrated Cronbach's alpha value of 0.89, suggesting good reliability across its items.

Harman's single-factor test was conducted to address common method bias. The test results revealed that the first factor explained 32% of the total variance, below the critical threshold of 50%. This indicates that common method bias is unlikely to significantly affect this study's results. The cross loadings of the study variables are given in Table 2.

5.2 Structural Analysis and Results

The series of tests were run to determine the descriptive analysis and construct correlations followed by the path estimates of the hypothesized relationships using Smart PLS-4. The results are discussed below:

Descriptive Analysis The descriptive analysis was conducted to provide preliminary valuable insights into the distribution and central tendencies of the study variables. The respondents from Pakistani manufacturing industries scored an average of 3.82 on the High-Performance Work Systems construct, indicating a moderate level of perceived high-performance work systems. On the contrary, the Potential Development construct scored a mean value of 4.16, reflecting significant positive perceptions of employees regarding the potential development in the sector. Training initiatives demonstrated a mean value of 3.7, reflecting a generally positive outlook toward training practices. AI construct depicted a mean value of 3.94, confirming employees' positive outlook towards adopting AI in their respective firms.

Table 1 Study constructs reliability scores

	Items	Loadings	Cronbach's alpha	rho_A	CR	AVE	VIF
High-Performance Work Systems	HPWS1	0.762	0.860	0.815	0.798	0.61	**1.837**
	HPWS2	0.816					**1.907**
	HPWS3	0.772					**1.775**
	HPWS4	0.712					**1.640**
	HPWS5	0.784					**1.183**
	HPWS6	0.715					**1.261**
	HPWS7	0.801					**1.129**
	HPWS8	0.829					**1.043**
	HPWS9	0.861					**1.246**
	HPWS10	0.712					**1.472**
	HPWS11	0.784					**1.489**
	HPWS12	0.715					**1.082**
	HPWS13	0.801					**1.000**
	HPWS14	0.829					**1.775**
	HPWS15	0.861					**1.727**
	HPWS16	0.712					**1.568**
	HPWS17	0.784					**1.393**
	HPWS18	0.715					**1.518**
	HPWS19	0.802					**1.033**
	HPWS20	0.821					**1.404**
Potential Development	PD1	0.817	0.812	0.818	0.767	0.663	**2.022**
	PD2	0.709					**1.613**
	PD3	0.767					**2.041**
	PD4	0.835					**1.428**
	PD5	0.812					**2.176**
Training Initiatives	TI1	0.716	0.823	0.893	0.788	0.718	**1.060**
	TI2	0.892					**2.098**
	TI3	0.811					**1.000**
	TI4	0.817					**1.000**
	TI5	0.883					**1.266**
Artificial Intelligence	AI1	0.819	0.897	0.845	0.812	0.734	**1.622**
	AI2	0.798					**1.512**
	AI3	0.89					**1.416**
	AI4	0.682					**7.094**
	AI5	0.907					**1.824**
	AI6	0.712					**1.850**
	AI7	0.723					**1.016**
	AI8	0.729					**1.000**
	AI9	0.801					**1.000**
	AI10	0.981					**1.356**

The descriptive findings (Table 3) offered an initial glimpse into the participants' perceptions. They established a basis for an in-depth inquiry into the hypothesized relationships among High-Performance Work Systems, Potential Development, Training initiatives, and Artificial Intelligence within the Pakistani manufacturing sector, followed by the correlations among the variables (Table 4).

Table 2 Cross-loadings

Indicator	AI Adoption	Employee Productivity	Potential Development	HPWS Practices	Training Initiatives
AI1	0.82	0.42	0.39	0.37	0.35
AI2	0.85	0.41	0.40	0.36	0.33
AI3	0.79	0.38	0.35	0.34	0.31
AI4	0.36	0.80	0.43	0.41	0.38
AI5	0.39	0.83	0.45	0.43	0.40
AI6	0.35	0.78	0.41	0.39	0.37
AI7	0.37	0.41	0.84	0.44	0.42
AI8	0.38	0.43	0.87	0.46	0.44
AI9	0.35	0.40	0.82	0.42	0.40
AI10	0.34	0.39	0.42	0.81	0.44
PD1	0.36	0.41	0.44	0.85	0.46
PD2	0.33	0.38	0.40	0.79	0.42
PD3	0.31	0.37	0.41	0.43	0.80
PD4	0.34	0.40	0.44	0.45	0.85
PD5	0.30	0.36	0.40	0.42	0.79
TI1	0.57	0.47	0.39	0.37	0.35
TI2	0.55	0.41	0.40	0.36	0.33
TI3	0.79	0.35	0.35	0.34	0.31
TI4	0.36	0.50	0.43	0.41	0.35
TI5	0.39	0.53	0.45	0.43	0.40
HPWS1	0.35	0.75	0.41	0.39	0.37
HPWS2	0.37	0.41	0.54	0.44	0.47
HPWS3	0.35	0.43	0.57	0.46	0.44
HPWS4	0.35	0.40	0.57	0.47	0.40
HPWS5	0.34	0.39	0.47	0.51	0.44
HPWS6	0.36	0.41	0.44	0.55	0.46
HPWS7	0.33	0.35	0.40	0.79	0.47
HPWS8	0.31	0.37	0.41	0.43	0.50
HPWS9	0.34	0.40	0.44	0.45	0.55
HPWS10	0.30	0.36	0.40	0.47	0.79
HPWS11	0.57	0.47	0.39	0.37	0.35
HPWS12	0.55	0.41	0.40	0.36	0.33
HPWS13	0.79	0.35	0.35	0.34	0.31
HPWS14	0.36	0.50	0.43	0.41	0.35
HPWS15	0.39	0.53	0.45	0.43	0.40
HPWS16	0.35	0.75	0.41	0.39	0.37
HPWS17	0.37	0.41	0.54	0.44	0.47
HPWS18	0.35	0.43	0.57	0.46	0.44
HPWS19	0.35	0.40	0.57	0.47	0.40
HPWS20	0.34	0.39	0.47	0.51	0.44

The findings (Table 4) confirmed that the correlation between high-work performance systems and potential development is 0.65, which shows a moderately strong relationship. The correlation between training initiatives and a high-work performance system is 0.42,

Table 3 Descriptive analyses

Study Variables	*N*	Minimum	Maximum	Mean	Standard Deviation
High-Performance Work Systems	208	1	5	3.82	0.85
Potential Development	208	1	5	4.16	0.62
Training initiatives	208	1	5	3.72	0.85
Artificial Intelligence	208	1	5	3.94	0.82

Table 4 Correlation analysis

Variable	HPWS	PD	TRAIN	AI
High-Performance Work Systems	1.00			
Potential Development	0.65	1.00		
Training initiatives	0.42	0.58	1.00	
Artificial Intelligence	0.53	0.47	0.41	1.00

Table 5 Fornell-Larcker Criterion

Constructs	AI	PD	TRAIN	HPWS
Artificial Intelligence	0.856			
Potential Development	0.853	0.846		
Training initiatives	0.802	0.787	0.837	
High-Performance Work Systems	0.787	0.869	0.869	0.862

which shows a weak and positive relationship. The correlation value of 0.53 between high-work performance systems and artificial intelligence shows a strong relationship.

Potential development and training initiatives have a strong relationship of 0.58, which indicates a positive relationship between them. Potential development has a weak relationship with artificial intelligence; its correlation value is $r = 0.47$. Training initiatives also have a weak and positive relationship with artificial intelligence.

Next, the Fornell-Larcker Criterion was used to evaluate the discriminant validity of constructs in a structural equation model, such as PLS-SEM. Discriminant validity assesses whether constructs measure distinct concepts separate from one another. The diagonal values in Table 5 represent the square root of the Average Variance Extracted (AVE) for each construct, indicating the amount of variance captured by the items within that construct relative to the total variance in the data. Higher diagonal values suggest stronger internal consistency within constructs. Off-diagonal values represent the correlations between constructs, with comparisons made to the diagonal values to determine discriminant validity. The results confirmed the discriminant validity of the constructs, as diagonal values generally exceed off-diagonal correlations. For instance, the correlation between AI and PD (0.853) is lower than the square roots of the AVE for both AI (0.856) and PD (0.853), indicating discriminant validity. This pattern also holds across other correlations, reinforcing confidence in the validity of the measurement model.

5.3 Structural Path Analysis & Hypotheses Testing

The structural path analysis and hypotheses testing yield insightful findings regarding the relationships among the study constructs. Beginning with Hypothesis 1, which examines the link between Artificial Intelligence (AI) and Potential Development, results indicate a significant positive association, suggesting that organizations utilizing AI technologies are likely to experience heightened potential development among employees. Similarly, Hypothesis 2 uncovers a significant positive relationship between AI and High-Performance Work Systems (HPWS), implying that AI adoption within organizations is correlated with the implementation of high-performance work systems (Murugesan et al. 2023).

Moving on to Hypothesis 3, which explores the connection between Potential Development and HPWS, findings reveal a significant positive association, indicating that organizations prioritizing employee potential development are more inclined to have high-performance work systems in place also supported by (Oruganti et al. 2023). Hypothesis 4 delves into the indirect relationship between AI and HPWS through the mediating role of potential development. The analysis underscores a substantial positive mediation effect, emphasizing the significance of potential development as a mechanism through which AI influences the establishment of high-performance work systems (Naeem et al. 2023).

Lastly, Hypothesis 5 examines the moderating effect of training initiatives on the relationship between AI and potential development. The results indicate a positive moderation effect, suggesting that training initiatives bolster the relationship between AI adoption and employee potential development (Ahmed et al. 2023). These findings underscore the intricate dynamics between AI adoption, potential development, training initiatives, and the cultivation of high-performance work environments within organizations, emphasizing the importance of strategic investments in AI and employee development initiatives (Fig. 2).

The results of the hypothesized relationships between the study constructs are discussed one after another. For detailed results, see Table 6.

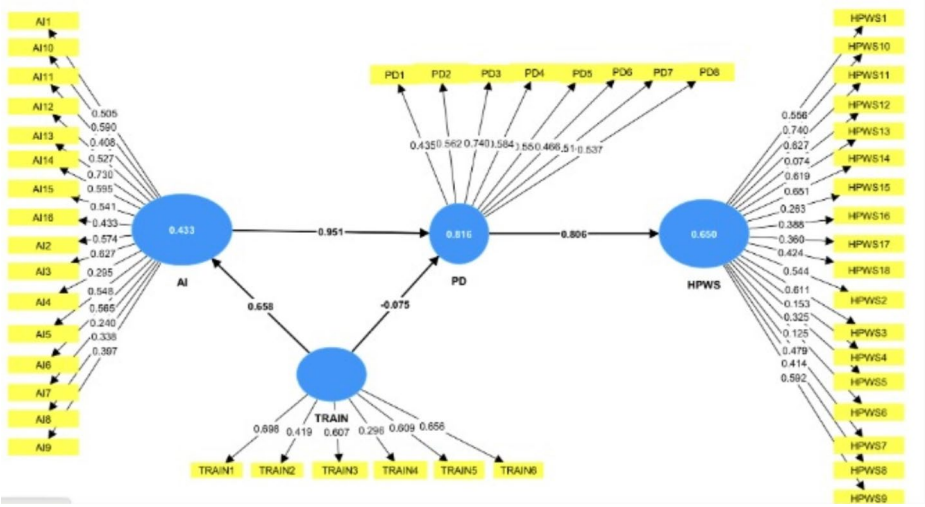

Fig. 2 Measurement Model

Table 6 Hypotheses testing results

		b-Value	t-Value	*p*-Value	Standard Error	LLCI	ULCI
H1	AI → PD	0.951	2.365	0.000	0.121	0.142	0.428
H2	AI → HPWS	0.456	3.210	0.000	0.142	0.187	0.825
H3	PD → HPWS	0.806	2.876	0.001	0.112	0.123	0.519
H4	AI → PD → HPWS	0.873	6.132	0.000	0.142	0.592	1.154
H5	TI * AI → PD	0. 658	3.201	0.002	0.130	0.162	0.672

Hypothesis 1 (H1): Artificial Intelligence → Potential Development: For H1, the coefficient is 0.951, the t-value is 2.365, and the p-value is 0.000. The standard error is 0.121, and the LLCI and ULCI are 0.142 and 0.428, respectively. These results suggest a significant positive link between Artificial Intelligence and Potential Development.

Hypothesis 2 (H2): Artificial Intelligence → High-Performance Work System: The coefficient (b-value) for H1 is 0.456, with a corresponding t-value of 3.210 and a p-value of 0.000. The standard error is 0.142, and the Lower and Upper Limits of the Confidence Interval (LLCI and ULCI) are 0.187 and 0.825, respectively. These values suggest a significant positive relationship between Artificial Intelligence and High-Performance Work Systems.

Hypothesis 3 (H3): Potential Development → High-Performance Work System: H3 demonstrates a coefficient of 0.806, a t-value of 2.876, and a p-value of 0.011. The standard error is 0.112, and the LLCI and ULCI are 0.123 and 0.519, respectively. This implies a significant positive association between Potential Development and High-Performance Work Systems.

Hypothesis 4 (H4): Artificial Intelligence → Potential Development → High-Performance Work System: H4 reveals a substantial coefficient of 0.873, a high t-value of 6.132, and a strongly significant p-value of 0.000 with indirect relationship. The standard error is 0.142, and the LLCI and ULCI are 0.592 and 1.154, respectively. This strong outcome indicates a significant positive mediation of potential development between Artificial Intelligence and High-Performance Work Systems and shows their indirect relationship.

Hypothesis 5 (H5): Training Initiatives * Artificial Intelligence → Potential Development: The coefficient for H5 is 0.658, with a t-value of 3.201 and a p-value of 0.002. The standard error is 0.130, and the LLCI and ULCI are 0.162 and 0.672, respectively. The values show the positive moderation of training initiatives between artificial intelligence and potential development, suggesting a meaningful positive moderation of training between Artificial Intelligence and Potential Development.

The results from the regression analysis underscore the significance and direction of the relationships examined in the study, offering valuable insights into the intricate interplay between Artificial Intelligence, Potential Development, Training initiatives, and High-Performance Work Systems within the manufacturing sector.

R-squared (R^2) The model accounts for 56.7% of the variance in High-Performance Work Systems, suggesting that each independent variable significantly improves high-performance work systems.

F-statistics F-statistics value of 21.816 with a p-value of 0.000 indicates the overall significance of the model. In contrast, its low p-value suggests that at least one independent variable significantly predicts its dependent variable. The p-value associated with the t-statistic is crucial for interpreting the results. It represents the probability of obtaining the observed results (or more extreme results) if there is no real relationship between the independent variables and the dependent variable. In other words, these findings demonstrate the relevance of artificial intelligence and potential development in improving high-performance work systems within manufacturing industries. Regression analysis provides concrete evidence for the relationships among variables, providing insight into individual and combined contributions (Suseno et al. 2022). Further examining these relationships may help organizations leverage AI strategically while expanding its potential development to create high-performing work systems.

6 Discussion

The current study examined whether applied artificial intelligence can support organizations' high-performance work systems through its potential development programs and training initiatives for human capital. The findings yielded valuable insights into the relationships among the applications of artificial intelligence (AI), employees' potential development (PD) and training initiatives, and high-performance work systems (HPWS) in the manufacturing sector (Alam and Lin 2022). These findings illuminate how these variables influence the most desired organizational outcomes of high-performance work systems (Adil and Ab Hamid 2020).

The study explored whether high-performance work systems can be improvised by using artificial intelligence (Martini et al. 2024). It was also imperative to determine the mediating role played by the employees' potential development programs in further strengthening the impact of artificial intelligence applications on HPWS; and if the training initiatives moderate the relationship between artificial intelligence and employees' potential development programs (Holm and Lorenz 2022). The significant positive relationships between these constructs and Artificial Intelligence adoption indicate their substantial influence. The positive relationship between HPWS and Artificial Intelligence adoption suggests that organizations investing in artificial intelligence technology-based practices are more likely to achieve successful high-performance work systems by upskilling their employees and preparing them for the future market (Pyzer-Knapp et al. 2022).

Raish and Krakowski (2021) conducted a quantitative study examining the relationship between AI augmentation and organizational performance. While their findings highlighted AI's positive impact on operational efficiency, our study expands upon this by exploring the specific mechanisms through which AI influences employee productivity. By employing PLS-SEM and explicitly addressing moderating factors, we offer a nuanced understanding

of how contextual variables shape the relationship between AI adoption and productivity outcomes (Patel et al. 2024).

Similarly, Smith et al. (2023) qualitatively analyzed AI integration strategies in large corporations. While their study provided valuable insights into organizational approaches to AI implementation, our research complements this by focusing on the actual outcomes of AI adoption, particularly its effects on employee productivity. By triangulating quantitative data with qualitative insights, we provide a comprehensive understanding of the multifaceted impacts of AI on organizational performance (Olan et al. 2022). Furthermore, our study adds to the existing literature by explicitly addressing methodological considerations such as common method bias and model fit estimation. By employing rigorous sampling techniques and robust statistical analyses, we enhance the credibility and reliability of our findings, contributing to methodological advancements in AI research (Mumali 2022).

The findings underscore the importance of investing in employee's potential development programs. Organizations that focus on enhancing the capabilities of their workforce are better positioned to embrace the transformative potential of AI (Sundaram and Zeid 2023). This finding echo that a well-prepared workforce is crucial for harnessing AI's benefits (Brynjolfsson & McAfee, 2017). The positive association between training initiatives and AI adoption emphasizes the value of structured training programs (Qi and Yao 2021). These programs equip employees with the skills required to effectively utilize AI tools and technologies, reducing resistance to adoption (Robbins et al., 2020).

The results of this study have practical implications for organizations in the manufacturing sector stated also by (Murugesan et al. 2023). To successfully implement high-performance work systems, firms should be willing to invest in AI technologies, with a particular focus on employee potential development, and establish comprehensive training initiatives to improve employees' day-to-day performance. These measures can enhance the workforce's readiness and increase their productivity (Andronie et al. 2021).

The study sheds light on the critical role of HR practices and employee development in adopting Artificial Intelligence in the manufacturing sector. The robustness of our measurement scales, as evidenced by reliability and validity assessments, strengthens the credibility of our findings (Banitaan et al. 2023). The practical implications highlight the importance of preparing the workforce for the AI-driven future, and avenues for future research have been identified to further advance our understanding of this dynamic relationship (Yang et al. 2021).

Artificial Intelligence and Potential Development have a significant relationship that underscores AI's essential role in supporting employee skill development and career advancement (Mohsin 2023). This finding aligns with earlier studies emphasizing AI's contribution to providing tailored learning opportunities, skill acquisition, and knowledge enhancement (Li et al. 2022). AI-powered training platforms or personalized learning experiences allow employees to acquire new competencies that ultimately contribute to their potential development (Behl et al. 2021).

The study findings confirming the association between AI and High-Performance Work Systems (HPWS), support the previous research emphasizing AI's transformative potential in optimizing organizational processes and increasing performance (Binsaeed et al. 2023). Integrating AI technologies may result in streamlining operations, improved decision-making, and resource allocation, all of which contribute to HPWS effectiveness and support the

idea that AI empowers organizations to implement sophisticated HR practices that elevate overall performance (Narbariya et al. 2022).

The significant correlation between Potential Development and High-Performance Work Systems (H3) evidences this relationship. Investing in employee growth and skill enhancement promotes an environment conducive to HPWS implementation (Pap et al. 2022); this aligns with Li et al. (2022), who state that nurturing employees' talents and aspirations not only increases individual capabilities but also helps create an optimal organizational culture (Li et al. 2022). Organizations prioritizing Potential Development are better poised to successfully apply HPWS practices (Aslam et al. 2023).

The study findings confirmed the mediation mechanism, suggesting that Potential Development mediates the relationship between AI and High-Performance Work Systems (Ari et al. 2020). The findings highlighted the employee growth's key role in connecting AI adoption and organizational performance (Khan and Liu 2023). As employees gain new skills through AI-enabled initiatives, they become better suited to contribute to implementing HPWS practices, creating an endless cycle of improved performance and development (Ayestaran et al., 2022).

Likewise, the moderation effect results suggested that training amplifies AI's positive impact on Potential Development (Li et al. 2022). This result is consistent with previous research indicating that well-designed training programs can boost AI adoption (Ari et al. 2020). Furthermore, AI training initiatives tailored specifically for employees' potential development can accelerate skill acquisition and competence building, further supporting employees' potential development (Zhou and Zou 2023).

One of the remarkable aspects of this research is its focus on cultivating employee potential. Innovation has become necessary within modern manufacturing environments; manufacturing firms that invest in nurturing creative and problem-solving capabilities within their workforce unlock an abundance of innovation potential (Byrne et al. 2021). By creating an environment in which employees are encouraged to think critically, offer suggestions, and experiment with AI-driven solutions, organizations can unlock the collective intelligence of their teams (Geng et al. 2023). This innovation extends beyond product development; it encompasses process optimization, supply chain management, customer engagement, and continuously finding more efficient methods (Plathottam et al. 2023). These findings also emphasize the significance of providing employees with the knowledge and skills required for navigating the AI era - this goes beyond providing job-specific training; organizations must build a flexible and adaptable workforce (Javaid et al. 2021).

As AI technologies advance and new tools emerge, organizations require employees who can rapidly learn and adapt. Training initiatives play a vital role here; they ensure employees remain relevant while welcoming AI-driven changes with open arms (Sullivan and Wamba 2024). An adaptable workforce doesn't fear technological shifts but embraces them as opportunities for personal and professional growth and development (Guo et al. 2023). This research highlights an essential principle: technology's effect on human capital is not a zero-sum game. Technology should not replace humans; instead, it should amplify human abilities. AI in manufacturing can handle repetitive tasks efficiently while providing insights. Humans provide creativity, critical thinking, and contextual knowledge.

Organizations that find the ideal balance between AI and human potential gain a competitive edge. By optimizing efficiency while maintaining human characteristics that drive inno-

vation and customer satisfaction, these businesses create a harmonious partnership between machines and minds, leading to greater performance results.

6.1 Study Implications

The findings demonstrate an intricate interaction among AI, Potential Development, Training and High-Performance Work Systems within the manufacturing sector. This research adds to a growing body of literature on integrating AI strategically into human resource practices, with particular attention paid to cultivating employee potential to maximize the benefits associated with its adoption. The theoretical implications arise from the mediation of PD between AI and HPWS, highlighting employee growth's centrality to translating AI-driven advancements into tangible organizational results. Furthermore, this research contributes to AI, HRM, and organizational behavior literature by providing empirical evidence of AI's complex pathways that shape workplace practices and performance.

Although this study focuses on Pakistan, its relevance extends far beyond. Manufacturing is at the core of economies worldwide and presents unique challenges and opportunities when adopting AI technologies. Organizations across borders can glean insights from this research to inform their strategies, with manufacturing firms that utilize it being better able to withstand disruptions, adjust when needed, and navigate toward future readiness. AI's true potential can only be unleashed with an innovative workforce of flexibility and skill. This shift does not just provide temporary relief; rather, it lays the groundwork for lasting success in an ever-evolving AI-dominated world.

Organizations within this sector may use these insights as part of comprehensive strategies designed to leverage technology-human capital synergies for improved performance and success in overall organizational operations. Employers in the manufacturing sector should leverage AI technologies to enhance employee performance management practices, particularly Potential Development (PD), that work seamlessly with High-Performance Work Systems (HPWS). Adopting this strategic approach requires extensive investments in AI-enabled training platforms and programs; by employing AI for performance management purposes, organizations can foster more agile and dynamic workforces, increasing overall productivity and excellence.

Training initiatives must be designed carefully to accompany the adoption of artificial intelligence technologies. Programs should stress developing skills compatible with AI technologies so employees can collaborate efficiently with these tools stated by (Martini et al. 2024). AI literacy, data analysis, and problem-solving abilities should form the cornerstone of training efforts. By aligning training efforts with AI adoption, organizations equip their workforces with competencies needed to unlock AI's full potential (Sundaram and Zeid 2023).

Organizations must recognize employee professional development (PD) as a strategic imperative. Bridging the gap between AI integration and organizational performance depends heavily on having a skilled, adaptable workforce that embraces continuous learning and growth opportunities (Murugesan et al. 2023). By prioritizing employee development initiatives as a priority, organizations will not only enhance performance but also bolster their competitive standing within digital ecosystems.

Promoting a culture of continual learning among employees is essential in today's era of ever-evolving AI technologies, where employees need to adapt quickly (Guo et al. 2023).

Organizations should provide multiple formal and informal opportunities for ongoing education to ensure a workforce that stays agile and responsive to technological advancement.

This research contributes to an evolving discourse regarding artificial intelligence's (AI's) effects on Human Resource Management (HRM) and organizational performance, particularly within manufacturing organizations (Patel et al. 2024). It highlights AI's transformative power while emphasizing its need for a comprehensive approach that maximizes its potential. Furthermore, the recommendations provide organizations looking to integrate AI effectively into their HR practices, navigate the digital era, and stay ahead in an ever-evolving manufacturing sector.

The study findings contributed to the academic literature on AI adoption and human resource management. They provided practical implications for manufacturing organizations looking forward to enhancing competitiveness through automation (Singh et al. 2020). It has implications for Pakistan's manufacturing sector, academia, and the broader business community (Ihsan et al. 2020). It offered invaluable insight to organizations striving to thrive in an AI-powered world by highlighting the critical role of Artificial Intelligence (AI) in creating High-Performance Work Systems (HPWS). Additionally, it expanded the understanding of potential development's mediating role and training's moderating influence in the AAI-HPWS relationship, further contributing to the theoretical discourse surrounding organizational behavior, technology adoption, and human resource management (Khan and Liu 2023). Manufacturers can use AI technologies to develop employee potential through personalized learning and skill acquisition, improving employee engagement, productivity, and organizational performance. Furthermore, this study stresses the significance of training initiatives aligned with AI adoption for optimizing employee development.

6.2 Conclusion, Limitations, & Future Directions

The research empirically investigates the interdependencies of AI, PD, Training initiatives, and HPWS within Pakistan's manufacturing sector. Its results demonstrate positive relationships among these constructs - AI's role in encouraging PD, which then increased HPWS. Training also magnified AI's effect on PD, reinforcing holistic approaches to technology integration and employee development. Finally, mediating between AI and HPWS provided an insightful view of its underlying mechanisms.

The positive relationships between these constructs shed light on AI's central role in supporting Personal Development (PD) and increasing HPWS effectiveness, thus heightening the effectiveness of HPWS. Furthermore, their findings illustrate its transformative potential as an enabler of employee growth and potential development. AI technologies integrate into organizational processes more seamlessly, increasing operational efficiency and fostering skilled and adaptable workers.

Furthermore, mediation analysis which exposes the role of PD between AI and HPWS, provides a key link in understanding how these variables interact to influence each other over time. This insight provides a roadmap for organizations seeking to leverage AI-driven Potential Development to implement High-Performance Work Systems. AI-enabled initiatives enable employees to hone their capabilities and ensure successful HPWS practices, ultimately enhancing sustained performance. Organizations can utilize this information to inform strategic decision-making regarding AI adoption, training program design, and employee development initiatives. With an integrated approach incorporating AI, PD,

Training initiatives, and HPWS mind, the manufacturing sector in Pakistan can create a productive ecosystem that leverages technological innovations while developing employees to their maximum potential (Kambur & Akar 2022).

Technology continues redefining industries, and this study illustrates human resource strategies' essential role in unlocking AI's full potential. It explores how AI interfaces with employee development and performance, emphasizing an adaptive, forward-looking approach. By responding appropriately to the challenges and opportunities presented by this interlinked network of variables, organizations can navigate rapidly changing dynamics within manufacturing sectors like theirs while maintaining sustained success through sustained technological change and human capital development (Rosunee & Unmar 2024).

Though this study provides insight into the relationships among AI, PD, Training initiatives, and HPWS, there is still plenty of room for further investigation. This research is poised to inspire further investigation of AI applications for Human Resource Management and their ethical considerations, thus furthering our knowledge of how technology and human behavior interact within organizational environments.

While this study has shed light on the intricate relationships among AI, Potential Development (PD), Training Initiatives, and High-Performance Work Systems (HPWS), it is important to acknowledge the limitation of cross-sectional data. Future research could benefit from longitudinal studies that span an extended period. Such studies would provide valuable insights into the causal directions of these relationships over time. Understanding how AI adoption influences PD, which in turn impacts HPWS and vice versa, can offer a more nuanced understanding of these dynamics and their evolution.

Various contextual factors influence organizational dynamics. This research provides a foundation for understanding the relationships, but future studies could delve deeper into the role of organizational culture and leadership. How do different organizational cultures impact the integration of AI into HR practices? How do leaders' attitudes and actions shape AI adoption and its impact on PD and HPWS? Exploring these contextual factors can provide richer insights into the complexities of AI-driven HR transformations.

This study focused on the manufacturing sector in Pakistan, and while the findings are valuable, there is an opportunity for broader comparative research. Examining AI integration in HR practices across different industries, such as healthcare, finance, or technology, can reveal sector-specific nuances and best practices. Similarly, comparative research across regions with varying levels of AI adoption can help identify global trends and regional variations in the relationship between AI, PD, Training Initiatives, and HPWS. As AI gains prominence in HR practices, ethical considerations become increasingly important. Future research could explore the ethical dimensions of AI adoption in HRM, addressing issues such as bias in AI algorithms, data privacy, transparency, and fairness. Additionally, examining AI governance frameworks and their impact on HR practices can guide organizations navigating the ethical complexities of AI.

While this research touched on AI's positive impact on employee Potential Development, future studies can delve deeper into its effects on employee well-being and job satisfaction. How does AI-driven HRM influence factors like work-life balance, job engagement, and employee satisfaction? Understanding AI's holistic impact on the workforce's quality of life is essential for responsible AI integration. Recent global events have highlighted the importance of crisis management. Future research could explore how AI-driven HR practices contribute to crisis preparedness and response.

Appendix 1: Questionnaire

Applied Artificial Intelligence

My organization has access to very large, unstructured, or fast-moving data for analysis.

My organization has the capacity to share data across business units and organizational boundaries.

My organization can prepare and cleanse AI data efficiently and assess data for errors.

My organization can obtain data at the right level of detail to produce meaningful insights.

My organization can anticipate resistance to change.

My organization can recognize the need to manage change.

My organization can communicate the reasons for change to its members.

My organization has a strong preference for high-risk projects with opportunities for very high returns.

I believe that the usage of new AI strengthens and harmonizes the skills of employees.

AI adds value to your organization's operations.

High-Performance Work Systems

In my organization, Employees are involved in job rotation.

In my organization, Employees are empowered to make decisions.

In my organization, jobs are designed according to employee skills and capabilities.

In my organization, the selection process is comprehensive, utilizing interviews and tests.

In my organization, the selection process emphasizes candidates' ability to contribute to the team.

In my organization, the selection process involves screening job candidates.

In my organization, the selection process prioritizes selecting the best candidate, regardless of the job.

In my organization, the selection process emphasizes promotion opportunities.

In my organization, the selection process prioritizes candidates' potential to learn, such as aptitude.

In my organization, training is conducted regularly.

In my organization, training programs are comprehensive.

In my organization, training programs are designed to develop firm-specific skills and knowledge.

In my organization, training programs emphasize on-the-job experience.

In my organization, performance evaluation is based on objective, quantifiable criteria.

In my organization, performance appraisals include feedback.

In my organization, incentives are based on team performance.

In my organization, compensation packages include extensive benefits.

In my opinion, the compensation in my organization includes high wages.

In my organization, the incentive system is associated with skill-based performance.

In my organization, compensation is related to individual performance.

Training Initiatives

My organization provides learning/training opportunities to meet the changing needs of the workplace.

In my organization training is encouraged and rewarded.

The on-the-job training receive by the employees is applicable to job.

The amount of training given to employees on the job is sufficient.

The training employees receive on the job meet organizational needs.

Potential Development

The organization provides opportunities for personal growth and development.

The opportunities provided by the organization helps to meet my developmental needs.

The organization provides resources and support for employees' skill enhancement and career progression.

I feel capable of acquiring new knowledge and skills to excel in my role.

I am motivated to take on challenges and embrace new responsibilities.

Author contributions All four Authors have contributed equally .Some of the areas are mentioned below. Conceptualization: Sajjad Zahoor, Shuili Yang, Iffat Sabir Chaudhry.Data Curation: Sajjad Zahoor, Iffat Sabir Chaudhry.Formal Analysis: Sajjad Zahoor, Iffat Sabir ChaudhryFunding Acquisition: Shuili Yang, Xiaoyan RenInvestigation: Shuili Yang, Iffat Sabir ChaudhryMethodology: Sajjad Zahoor, Iffat Sabir ChaudhryProject Administration: Shuili Yang, Xiaoyan RenResources: Shuili Yang, Iffat Sabir ChaudhrySoftware: Sajjad ZahoorSupervision: Shuili YangValidation: Sajjad Zahoor, Iffat Sabir ChaudhryVisualization: Shuili Yang, Xiaoyan RenWriting-original draft: Sajjad Zahoor Writing-review & editing: Iffat Sabir Chaudhry, Xiaoyan Ren.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

Informed consent Informed consent was obtained from all subjects involved in the study.

Institutional review board statement Ethical approval was not required for this study as participant's consentswas sought before participating in the survey and Participants don't belong to any vulnerable groups

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